CENTRAL UNIVERSITY OF SOUTH BIHAR
Department of Computer Science

End Term Examination **Academic Session: 2025-26** **Semester: 1st**

Programme: M.Sc. in Artificial Intelligence Duration: 3 hours
Course Code: CAI81DE00904 Course Title: Design & Analysis of Algorithms
Enrolment No. (To be filed by student): CUSB2502332012 Total Points: 70

There are three sections (A, B, and C). Answer all questions as per instructions. Points against each question are mentioned in the parenthesis.

Section – A

Answer all the questions in not more than 50 words. Each question carries 1Point.

(1 x 10 = 10 Points)

1. Explain NP-class problems.
2. State Cook's theorem.
3. Explain linear probing in hashing.
4. Define minimum spanning tree.
5. Write only names of standard methods for designing an algorithm.
6. Solve the recurrence relation $T(n) = 3T(n/2) + n^2$ using the master theorem.
7. When does a collision occur in hashing?
8. Define big θ-notation for a given function g(n).
9. Explain NP- hard problem.
10. Write main characteristics of the algorithm.

Section – B

Answer ANY FIVE questions in not more than 300 words. Each question carries 6 Points.

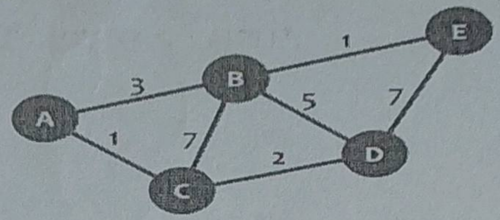
(6 x 5 = 30 Points)

1. Find the time complexity of the following recurrence relation:

$$T(n) = T(n/2) + 1, \text{ when } n \geq 2, \text{ and}$$
$$T(1) = 1.$$

   Use substitution method for solving the above recurrence relation.
2. Insert integer keys 8, 25, 5, 10 and 4 into an initially empty AVL tree. Show each step clearly.
3. Discuss three ways to analyze time complexity of an algorithm.

4. Insert items with keys: 89, 18, 49, 58 and 9 into an empty hash table of size 10. Hash function is hash(x) = x mod 10. Use quadratic hashing techniques for collision resolution.
5. Write the recurrence relation for quick sort in the worst case time complexity analysis. Also, solve the recurrence relation. **(2+ 4 marks)**
6. Compute the shortest path between the nodes C and E using Dijkstra's algorithm for the following graph:



Section – C

Answer ANY TWO questions in not more than 1000 words. Each question carries 15 Points.
(15 x 2 = 30 Points)

1. Build a maxheap H from the following list of numbers: 40, 35, 55, 45, 65, 80, 25 and 95. Clearly draw the maxheap after each insertion. Also, find the maximum number of the above list by deleting the root of the maxheap H. **(12+3 marks)**
2. We want to encode a text with the characters a, b, c, d, e, f, and g occurring with the following frequencies (in thousands):

Char	a	b	c	d	e	f	g
Frequency	45	13	12	16	9	15	5

Now, perform the following:
(a) Build the Huffman tree. **(7marks)**
(b) Find the weighted path length. **(3 marks)**
(c) Find the Huffman coding for above characters. **(5 marks)**

3. Find minimum spanning tree of the following graph using Kruskal and Prim's algorithms.